



Diagram illustrating a quantum circuit transformation:

The circuit starts with a horizontal line entering a square box labeled $\hat{U}$. A horizontal line exits the box to the right, followed by a double-headed arrow $\longleftrightarrow$.

The transformation is defined by the following equation:

$$\hat{U} = \frac{1}{\sqrt{d}} \sum_{j,k=0}^{d-1} \omega_d^{jk} |j\rangle \langle k|, \quad \omega_d = e^{\frac{2\pi i}{d}}$$